From State Space to Decision Space

A General Formal Theory of Geometry-Native Computation Under Constraints and Finite Resources

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Abstract

We formalize the passage from a representation of states to a representation of decision-relevant distinctions. Let a state space carry an equivalence relation induced by the task, constraints, evidence, and available actions. The quotient by task-indistinguishability defines a decision space in which geometrically distinct states may become equivalent, while states that are close under a physical or computational metric may remain decision-distinct. We define a geometry-native computation as one whose admissible transformations act directly on this task-relative quotient rather than merely approximating an externally specified numerical objective. The central result is a quotient-consistency theorem: if a transformation respects the task equivalence relation and preserves all protected predicates, then it induces a well-defined map on decision classes; conversely, failure of congruence yields ambiguity in the decision semantics. We characterize when compression is lossless with respect to a governed decision contract and when representation reduction necessarily destroys a distinction required for feasibility, identity, or certification. Finite computation is modeled through a budget vector covering representation, search, computation, verification, proof, and stopping. Randomization may generate candidates, but does not alter the objective, feasible set, or certification relation.

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1 Introduction

1.1 The two-space problem

Adaptive systems operate on states. Decisions are made about distinctions. These are not the same thing, and the difference is often the source of silent error.

A state space is a set of configurations, measurements, or representations. A decision space is a set of equivalence classes of states that cannot be distinguished by the task at hand. Two states that are geometrically far apart may be decision-equivalent if the task cannot tell them apart. Two states that are metrically close may be decision-distinct if the task requires a distinction the metric does not see.

The passage from state space to decision space is therefore a quotient. The quotient is not geometric: it is induced by the task, the constraints, the evidence, and the available actions. A representation that ignores this quotient may preserve geometric structure while destroying decision structure, or vice versa.

This paper formalizes the passage, defines what it means for a computation to be *geometry-native* with respect to the quotient, proves a quotient-consistency theorem for admissible transformations, characterizes lossless compression with respect to a governed decision contract, and models finite computation through a budget vector.

1.2 Relation to the constitutional framework

The companion paper *Constitutional Geometry of Adaptive Systems* established that admissibility is a structural property of the state space, that constitutional transitions are partial maps on the admissible set, and that protected invariants compose under sequential admissibility. The present paper addresses the next question: once a state space is given, how do we pass to the space of decisions, and what does it mean for a computation to act on that space directly?

The two papers are complementary. The first is about what states are permitted. The second is about what states are distinguished.

1.3 Contributions

1. We define task-induced equivalence and the decision quotient, and characterize the induced decision relation on the quotient.

2. We define geometry-native computation and show that it is exactly computation whose transformations act on the quotient.

3. We prove the quotient-consistency theorem: a transformation respecting the task equivalence relation and preserving protected predicates induces a well-defined map on decision classes, and failure of congruence yields ambiguity.

4. We characterize lossless compression with respect to a governed decision contract and identify necessary losses.

5. We model finite computation through a budget vector covering representation, search, computation, verification, proof, and stopping.

6. We show that randomization may generate candidates but does not alter the objective, feasible set, or certification relation.

1.4 Falsification strategy

Every claim is falsifiable by a structural counterexample. A transformation is claimed to be quotient-consistent; a counterexample is a transformation that respects the task equivalence relation on some pair but fails to induce a well-defined map on decision classes. A compression is claimed lossless; a counterexample is a pair of states identified by the compression whose decision roles differ. A budget vector is claimed complete; a counterexample is a computation whose cost exceeds the vector while producing a certified decision.

2 State Space, Task, and Equivalence

2.1 State space

Definition 2.1 (State space)**.** *A* state space *is a set X equipped with:*

- *a family of* task predicates $\{T_k : X \to \{0,1\}\}_{k\in K}$*;*
- *a family of* constraints $\{C_j : X \to \{0,1\}\}_{j\in J}$*;*
- *an* evidence structure $: X \to \mathcal{E}$*;*
- *a family of* available actions $\{A_i : X \rightharpoonup X\}_{i\in I}$*.*

The governed problem *on X is the tuple $\mathcal{P} = (X, \{T_k\}, \{C_j\}, , \{A_i\})$.*

The state space is not merely a set. It carries the structure that makes decisions possible: tasks, constraints, evidence, and actions.

2.2 Decision-relevant features

Definition 2.2 (Decision-relevant feature)**.** *A* decision-relevant feature *is any function $f : X \to V$ such that the decision induced by the governed problem depends on f. The* decision signature *of a state x is the tuple*

$$\sigma(x) = \big(\{T_k(x)\}_k, \{C_j(x)\}_j, (x), \{A_i(x)\}_i\big)$$

whenever this tuple is well-defined.

The decision signature records everything about a state that the governed problem can observe.

2.3 Task-induced equivalence

Definition 2.3 (Task-induced equivalence)**.** *Two states $x, y \in X$ are* task-equivalent*, written $x \sim y$, if they have the same decision signature:*

$$x \sim y \iff \sigma(x) = \sigma(y).$$

Task-equivalence is the indistinguishability relation of the governed problem. It is the finest equivalence the task can resolve.

Proposition 2.4. *$\sim$ is an equivalence relation on X.*

Proof. Reflexivity, symmetry, and transitivity follow from the corresponding properties of equality on decision signatures. □

Remark 2.5. *The definition of $\sim$ is task-relative. Different governed problems induce different equivalences on the same underlying state space. There is no task-free notion of state equivalence.*

2.4 Physical and computational metrics

The state space may also carry a metric or a bilinear discernability form.

Definition 2.6 (Discernability form)**.** *A* discernability form *is a symmetric bilinear map*

$$B : X \times X \to \mathbb{R}$$

satisfying $B(x, x) \geq 0$ *for all* $x \in X$*. The* discernability *of* x *and* y *is* $B(x - y, x - y)$ *when* X *is a vector space, or* $B(x, y)$ *when* X *is a general set with a kernel.*

Definition 2.7 (Metric)**.** *A* metric *on* X *is a function* $d : X \times X \to [0, \infty)$ *satisfying the standard axioms of identity, symmetry, and the triangle inequality.*

The metric and the discernability form capture physical or computational proximity. Task-equivalence captures decision proximity. These are generally independent.

Proposition 2.8 (Independence of task-equivalence and metric proximity)**.** *In general:*

1. $x \sim y$ *does not imply* $d(x, y) = 0$ *or* $B(x - y, x - y) = 0$*.*
2. $d(x, y) = 0$ *or* $B(x - y, x - y) = 0$ *does not imply* $x \sim y$*.*

Proof. For (1): consider $X = \mathbb{R}^2$ with Euclidean metric, task predicate $T(x_1, x_2) = 1$ iff $x_1 > 0$, and no constraints, evidence, or actions. Then $(1, 0) \sim (1, 10^6)$ but $d = 10^6$. For (2): consider $X = \mathbb{R}^2$ with Euclidean metric, task predicate $T(x_1, x_2) = 1$ iff $x_1 = x_2$, and no constraints. Then $(1, 1)$ and $(1, 1 + 10^{-100})$ are metrically almost identical but task-distinct. □

Remark 2.9. *The proposition formalizes the abstract's observation: geometrically distinct states may become equivalent, and metrically close states may remain decision-distinct.*

3 The Decision Quotient

3.1 Quotient construction

Definition 3.1 (Decision space)**.** *The* decision space *is the quotient*

$$\mathcal{D} = X/\sim .$$

An element $[x] \in \mathcal{D}$ *is a* decision class*. The* canonical projection *is* $\pi : X \to \mathcal{D}$*,* $\pi(x) = [x]$*.*

The decision space is the set of distinctions the task can make. It is generally much smaller than X: many geometrically distinct states collapse to a single decision class.

3.2 Induced structure on the quotient

Definition 3.2 (Induced decision relation)**.** *Define the* decision relation $\succeq_{\mathcal{D}}$ *on* $\mathcal{D}$ *by*

$$[x] \succeq_{\mathcal{D}} [y] \iff \text{there exist } x' \in [x], y' \in [y] \text{ with } x' \succeq_c y',$$

where $\succeq_c$ *is the constitutional decision relation on* X*.*

Proposition 3.3. *If* $\succeq_c$ *is constant on* $\sim$*-classes, then* $\succeq_{\mathcal{D}}$ *is well-defined and independent of representatives.*

Proof. Suppose $x_1 \sim x_2$ and $y_1 \sim y_2$. We must show $x_1 \succeq_c y_1 \iff x_2 \succeq_c y_2$. By hypothesis, $x_1 \succeq_c y_1 \iff x_2 \succeq_c y_1 \iff x_2 \succeq_c y_2$. □

The proposition shows that the decision relation descends to the quotient exactly when the constitutional relation is invariant under task-equivalence. This is the first instance of a quotient-consistency condition.

Definition 3.4 (Governed decision contract)**.** *A* governed decision contract *on X is a tuple*

$$\mathcal{C} = (\sim, \succeq_c, \mathcal{A}, \mathcal{I}, \mathcal{W})$$

where $\sim$ is the task equivalence, $\succeq_c$ is the constitutional decision relation, $\mathcal{A} \subseteq X$ is the admissible set, $\mathcal{I}$ is a protected invariant family, and $\mathcal{W}$ is a set of valid transition witnesses.

The contract packages the decision-relevant structure. A computation is *contract-respecting* if it does not alter any component of $\mathcal{C}$.

3.3 Decision maps

Definition 3.5 (Decision map)**.** *A* decision map *is a function $\Delta : \mathcal{D} \to \mathcal{O}$ from decision classes to outcomes. An outcome space $\mathcal{O}$ is any set with a preorder $\succeq_{\mathcal{O}}$.*

Definition 3.6 (Lift of a decision map)**.** *A* lift *of Δ to X is a function $f : X \to \mathcal{O}$ such that $f = \Delta \circ \pi$, i.e., $f(x) = \Delta([x])$ for all x.*

The decision map is the task-level object. The lift is its representation-level realization. The quotient-consistency question is whether a given f on X is a lift of some Δ on $\mathcal{D}$.

4 Geometry-Native Computation

4.1 Definition

Definition 4.1 (Geometry-native computation)**.** *A computation $\mathcal{K}$ on X is* geometry-native with respect to the contract $\mathcal{C}$ *if every admissible transformation δ of $\mathcal{K}$ satisfies:*

1. *δ is a partial map on X respecting the admissible set $\mathcal{A}$;*
2. *δ respects the task equivalence: $x \sim y \implies \delta(x) \sim \delta(y)$ whenever both are defined;*
3. *δ preserves the protected invariants $\mathcal{I}$;*
4. *δ carries valid witnesses in $\mathcal{W}$.*

Geometry-native computation is not computation on X that happens to be useful. It is computation whose admissible transformations factor through the decision quotient.

Proposition 4.2 (Factorization)**.** *If $\delta : X \rightharpoonup X$ respects $\sim$, then there exists a unique partial map $\bar{\delta} : \mathcal{D} \rightharpoonup \mathcal{D}$ such that $\bar{\delta} \circ \pi = \pi \circ \delta$ on $\mathrm{dom}(\delta)$.*

Proof. Define $\bar{\delta}([x]) = [\delta(x)]$ for $x \in \mathrm{dom}(\delta)$. Well-definedness follows from $\sim$-respect: if $x \sim y$, then $\delta(x) \sim \delta(y)$, so $[\delta(x)] = [\delta(y)]$. Uniqueness follows from surjectivity of π. □

Remark 4.3. *The factorization is the formal content of the abstract's claim that geometry-native computation acts directly on the task-relative quotient rather than approximating an externally specified objective.*

4.2 Contrast with objective-approximating computation

An *objective-approximating computation* acts on X to maximize a scalar function $J : X \to \mathbb{R}$. It does not, in general, respect $\sim$. Two task-equivalent states may have different J values, and the computation may distinguish them unnecessarily. The result is effort spent on distinctions the task cannot use.

Example 4.4. *Let $X = \mathbb{R}^2$ with task predicate $T(x_1, x_2) = 1$ iff $x_1 > 0$ and no other structure. Let $J(x_1, x_2) = x_2$. Then all states with $x_1 > 0$ are task-equivalent, but the objective varies with x_2. An objective-approximating computation will search over x_2 to maximize J, spending resources on a distinction the task cannot resolve. A geometry-native computation recognizes that the decision space is $\{+, -\}$ and acts only on the sign.*

5 Quotient-Consistency

5.1 Statement

Theorem 5.1 (Quotient-consistency)**.** *Let $\delta : X \rightharpoonup X$ be a partial map. The following are equivalent:*

1. *δ respects the task equivalence $\sim$: for all $x, y \in \mathrm{dom}(\delta)$, $x \sim y \implies \delta(x) \sim \delta(y)$.*
2. *δ induces a well-defined partial map $\bar{\delta} : \mathcal{D} \rightharpoonup \mathcal{D}$ with $\bar{\delta} \circ \pi = \pi \circ \delta$ on $\mathrm{dom}(\delta)$.*
3. *δ is constant on $\sim$-classes up to $\sim$-equivalence: for every $\sim$-class $[x] \subseteq \mathrm{dom}(\delta)$, the image $\delta([x])$ lies in a single $\sim$-class.*

Proof. (1) $\implies$ (2): Proposition 4.2. (2) $\implies$ (3): If $\bar{\delta}$ is well-defined, then for $x, y \in [x]$, $\pi(\delta(x)) = \bar{\delta}([x]) = \pi(\delta(y))$, so $\delta(x) \sim \delta(y)$. (3) $\implies$ (1): Immediate. □

Corollary 5.2 (Failure of congruence)**.** *If δ does not respect $\sim$, then there exist $x \sim y$ with $\delta(x) \not\sim \delta(y)$. The induced map on $\mathcal{D}$ is then ambiguous: the value of $\bar{\delta}([x])$ depends on the choice of representative.*

Proof. By definition, if δ does not respect $\sim$, such x, y exist. If we attempt to define $\bar{\delta}([x]) = [\delta(x)]$, then $\bar{\delta}([x]) = [\delta(y)]$ is a different class, and the definition is not well-defined. □

5.2 Preservation of protected predicates

Theorem 5.3 (Quotient-consistency with invariants)**.** *Let $\delta : X \rightharpoonup X$ be a partial map that respects $\sim$ and preserves a protected invariant family $\mathcal{I}$. Then δ induces a partial map $\bar{\delta} : \mathcal{D} \rightharpoonup \mathcal{D}$ that preserves the induced invariant family $\bar{\mathcal{I}}$ on $\mathcal{D}$.*

Proof. By Theorem 5.1, δ induces $\bar{\delta}$. For the invariant preservation, define $\bar{I}_j([x]) = I_j(x)$ for each $I_j \in \mathcal{I}$. This is well-defined because $\sim$-equivalent states have the same decision signature, which includes the values of protected predicates. Then for $[x] \in \mathrm{dom}(\bar{\delta})$,

$$\bar{I}_j(\bar{\delta}([x])) = I_j(\delta(x)) = I_j(x) = \bar{I}_j([x]).$$

□

5.3 Necessity of each hypothesis

Proposition 5.4. *Both hypotheses of Theorem 5.3 are necessary.*

Proof. Without $\sim$-respect, Corollary 5.2 shows the induced map is ambiguous. Without invariant preservation, the induced map is well-defined but does not preserve $\bar{\mathcal{I}}$: for $[x]$ with $\bar{I}_j(\bar{\delta}([x])) \neq \bar{I}_j([x])$, the preservation fails. □

6 Lossless Compression

6.1 Compression maps

Definition 6.1 (Compression)**.** *A* compression *is a surjective map $R : X \to Z$ where Z is a representation space. The* fibers *of R are the sets $R^{-1}(z)$ for $z \in Z$.*

Compression reduces representation size. The question is whether it preserves decision structure.

Definition 6.2 (Lossless compression for a decision contract)**.** *A compression $R : X \to Z$ is* lossless *with respect to a governed decision contract $\mathcal{C}$ if every decision map $\Delta : \mathcal{D} \to \mathcal{O}$ on the decision space factors through R: there exists $\tilde{\Delta} : Z \to \mathcal{O}$ with $\Delta \circ \pi = \tilde{\Delta} \circ R$.*

Theorem 6.3 (Lossless compression criterion)**.** *A compression R is lossless for $\mathcal{C}$ iff for every $z \in Z$, all states in the fiber $R^{-1}(z)$ are task-equivalent:*

$$R(x) = R(y) \implies x \sim y.$$

Proof. ($\Leftarrow$) Suppose $R(x) = R(y) \implies x \sim y$. Then π factors through R: define $\bar{\pi} : Z \to \mathcal{D}$ by $\bar{\pi}(z) = [x]$ for any $x \in R^{-1}(z)$. This is well-defined by hypothesis. Then any Δ on $\mathcal{D}$ gives $\tilde{\Delta} = \Delta \circ \bar{\pi}$ on Z.

($\Rightarrow$) Suppose R is lossless. Then π factors through R: there exists $\bar{\pi} : Z \to \mathcal{D}$ with $\pi = \bar{\pi} \circ R$. If $R(x) = R(y)$, then $\pi(x) = \bar{\pi}(R(x)) = \bar{\pi}(R(y)) = \pi(y)$, so $x \sim y$. □

Corollary 6.4. *A compression is lossless for $\mathcal{C}$ iff every fiber of R is contained in a single $\sim$-class.*

Corollary 6.5. *If R is lossless and $\delta : X \rightharpoonup X$ is a geometry-native transformation, then δ descends to Z: there exists $\tilde{\delta} : Z \rightharpoonup Z$ with $\tilde{\delta} \circ R = R \circ \delta$.*

Proof. Since R is lossless, π factors as $\bar{\pi} \circ R$. Since δ respects $\sim$, it descends to $\bar{\delta}$ on $\mathcal{D}$. Define $\tilde{\delta}(z) = R(\delta(x))$ for $x \in R^{-1}(z)$. Well-definedness: if $R(x) = R(x')$, then $x \sim x'$ by Theorem 6.3, so $\delta(x) \sim \delta(x')$, hence $R(\delta(x)) = R(\delta(x'))$ by the lossless property again. □

6.2 Necessary losses

Theorem 6.6 (Necessary loss)**.** *If R identifies two states x, y with $x \not\sim y$, then the compression destroys a decision-relevant distinction. Specifically, there exist a decision contract $\mathcal{C}$ and a decision map Δ on $\mathcal{D}$ that does not factor through R.*

Proof. Since $x \not\sim y$, we have $[x] \neq [y]$ in $\mathcal{D}$. Choose $\mathcal{O} = \{0, 1\}$ with $0 \prec 1$ and define $\Delta([x]) = 1$, $\Delta([y]) = 0$. Suppose Δ factors through R: $\Delta \circ \pi = \tilde{\Delta} \circ R$ for some $\tilde{\Delta}$. Then $\tilde{\Delta}(R(x)) = \Delta(\pi(x)) = 1$ and $\tilde{\Delta}(R(y)) = \Delta(\pi(y)) = 0$. But $R(x) = R(y)$, contradiction. □

Remark 6.7. *The theorem formalizes the abstract's claim that representation reduction necessarily destroys a distinction required for feasibility, identity, or certification when the reduction collapses task-distinct states. The three cases correspond to three choices of decision map: feasibility (admissibility status differs), identity (identity classes differ), certification (certificate status differs).*

Corollary 6.8 (Feasibility loss)**.** *If $R(x) = R(y)$ with $x \in \mathcal{A}$ and $y \notin \mathcal{A}$, then R is not lossless for any contract $\mathcal{C}$ that includes the admissible set $\mathcal{A}$.*

Proof. Choose the decision map $\Delta([x]) = 1$ iff $[x] \subseteq \mathcal{A}$. By Theorem 6.6, Δ does not factor through R. □

Corollary 6.9 (Identity loss)**.** *If $R(x) = R(y)$ with x, y in different identity classes, then R is not lossless for any contract that includes identity.*

Corollary 6.10 (Certification loss)**.** *If $R(x) = R(y)$ with x certified and y uncertified, then R is not lossless for any contract that includes certification.*

6.3 The tradeoff

Proposition 6.11 (Lossless compression bound)**.** *The finest lossless compression for $\mathcal{C}$ has fibers exactly the $\sim$-classes. Any coarser compression destroys decision structure.*

Proof. By Theorem 6.3, lossless compressions have fibers contained in $\sim$-classes. The coarsest such is $R = \pi$, whose fibers are exactly the $\sim$-classes. Any coarser R identifies states in different $\sim$-classes, so by Theorem 6.6 it is not lossless. □

Remark 6.12. *The proposition says that the decision quotient $\mathcal{D} = X/\sim$ is the unique finest lossless compression. It is the canonical decision representation. Any coarser representation trades decision fidelity for compactness. This is the fundamental compression tradeoff for governed decision contracts.*

7 Finite Computation

7.1 The budget vector

Definition 7.1 (Resource budget)**.** *A* resource budget *is a vector*

$$B = (B_{rep}, B_{search}, B_{comp}, B_{ver}, B_{proof}, B_{stop})$$

of finite, nonnegative resource limits corresponding to representation, search, computation, verification, proof, and stopping.

The six components correspond to six distinct resource demands of a governed computation.

Representation. The cost of encoding, storing, and accessing the state representation.

Search. The cost of exploring candidates.

Computation. The cost of evaluating functions, transitions, and invariants.

Verification. The cost of checking that a candidate satisfies the constraints and the contract.

Proof. The cost of constructing or checking a certificate of optimality, feasibility, or invariant preservation.

Stopping. The cost of deciding when to terminate. Stopping is not free: it requires evidence that continued search cannot change the certified decision.

7.2 The budget-constrained computation

Definition 7.2 (Budget-constrained computation)**.** *A* budget-constrained computation *on X is a computation $\mathcal{K}$ together with an allocation*

$$b = (b_{rep}, b_{search}, b_{comp}, b_{ver}, b_{proof}, b_{stop})$$

with $0 \leq b_i \leq B_i$ for each component. The computation is feasible *if each component's actual consumption is at most b_i.*

Theorem 7.3 (Budget completeness)**.** *If a computation consumes a resource not accounted for in B, then its certified output cannot be compared with computations constrained by B.*

Proof. Suppose computation $\mathcal{K}_1$ uses resource $r \notin B$ and produces certified output o_1. Suppose computation $\mathcal{K}_2$ is constrained by B and produces certified output o_2. The certification of o_1 depends on r, which is unconstrained in B. The certification of o_2 depends only on resources in B. The two certifications are not commensurable: no comparison of o_1 and o_2 at the level of certified decisions is well-defined. □

Remark 7.4. *The theorem formalizes the abstract's claim that finite computation is modeled through the budget vector. Omitting a resource from the vector is not neutral: it makes the certified decision incommensurable with decisions that respect the vector.*

7.3 Budget for geometry-native computation

Proposition 7.5 (Geometry-native budget advantage)**.** *Let $\mathcal{K}$ be a geometry-native computation on X with contract $\mathcal{C}$, and let $\mathcal{K}'$ be an objective-approximating computation on X with the same contract. Then $\mathcal{K}$ can be realized with a strictly smaller search budget than $\mathcal{K}'$ whenever the decision quotient is a proper quotient of X.*

Proof. Since $\mathcal{K}$ acts on $\mathcal{D} = X/\sim$ while $\mathcal{K}'$ acts on X, and $|\mathcal{D}| < |X|$ when the quotient is proper, the search space for $\mathcal{K}$ is smaller. Formally, any search over X can be replaced by a search over $\mathcal{D}$ by the factorization property (Proposition 4.2), with a corresponding reduction in the number of candidates examined. □

Remark 7.6. *The proposition gives the formal reason geometry-native computation is efficient: it does not waste budget on distinctions the task cannot resolve.*

8 Randomization

8.1 Randomization as candidate generation

Definition 8.1 (Randomized computation)**.** *A* randomized computation *is a computation whose transition selection depends on a random source. A* randomized candidate generator *is a randomized computation whose output is a set of candidate states.*

Definition 8.2 (Objective, feasible set, certification relation)**.** *For a governed decision contract $\mathcal{C}$, the* objective J*, the* feasible set $\mathcal{A}$*, and the* certification relation $\mathcal{V} \subseteq X \times X$ *are fixed by the contract and are not altered by randomization.*

Theorem 8.3 (Randomization does not alter the contract)**.** *Let $\mathcal{K}$ be a randomized candidate generator for a contract $\mathcal{C}$. Then for every execution of $\mathcal{K}$, the objective J, the feasible set $\mathcal{A}$, and the certification relation $\mathcal{V}$ are unchanged.*

Proof. By definition, $\mathcal{K}$ outputs candidates. It does not modify J, $\mathcal{A}$, or $\mathcal{V}$, which are part of the contract $\mathcal{C}$ and are not among the outputs of $\mathcal{K}$. Therefore they are unchanged. □

Corollary 8.4. *Randomization can change which candidates are found, but cannot change which candidates are certified or which decisions are valid.*

Remark 8.5. *The corollary formalizes the abstract's claim that randomization may generate candidates, but does not alter the objective, feasible set, or certification relation. This is important: randomization is a search heuristic, not a decision mechanism.*

8.2 Interaction with the budget

Proposition 8.6. *A randomized candidate generator consumes search budget without guaranteeing coverage of the decision space.*

Proof. A randomized generator samples from a distribution on X. Coverage of $\mathcal{D}$ requires either exhaustive search or a certificate of coverage. Randomization provides neither without additional structure (e.g., ergodicity, mixing time bounds). □

Remark 8.7. *Randomization is therefore best understood as a search-budget allocation strategy, not as a decision procedure. It may be efficient in practice; it does not alter the constitutional structure.*

9 Falsifiers and Structural Counterexamples

9.1 Falsifier for quotient-consistency

A counterexample to Theorem 5.1 would be a transformation δ that respects $\sim$ on all pairs but does not induce a well-defined map on $\mathcal{D}$. No such example exists: the theorem is a direct consequence of the universal property of quotients. A counterexample to the *necessity* of $\sim$-respect would be a transformation that does not respect $\sim$ but still induces a well-defined map. This is impossible by Corollary 5.2.

9.2 Falsifier for lossless compression

A counterexample to Theorem 6.3 would be a compression R that identifies task-distinct states but still admits a factorization of every decision map through R. By Theorem 6.6, such an R cannot exist: the decision map Δ that separates the identified classes does not factor.

9.3 Falsifier for budget completeness

A counterexample to Theorem 7.3 would be a computation that consumes an unaccounted resource but still produces a certified output commensurable with B-constrained computations. Such an example would require the unaccounted resource to be non-essential to the certification, which by definition contradicts its consumption.

9.4 Falsifier for randomization neutrality

A counterexample to Theorem 8.3 would be a randomized computation that alters J, $\mathcal{A}$, or $\mathcal{V}$. This would require the randomization to modify the contract, which is outside the definition of a candidate generator.

9.5 How to test the theory

1. Specify X, the task predicates, the constraints, the evidence structure, and the available actions.
2. Compute the task-equivalence $\sim$ and the decision quotient $\mathcal{D}$.
3. Verify that the constitutional decision relation is constant on $\sim$-classes.
4. For each admissible transformation, check $\sim$-respect and invariant preservation.
5. For each compression, check whether fibers are contained in $\sim$-classes.
6. For each computation, allocate resources according to the budget vector and verify feasibility.
7. For each randomized generator, verify that the contract is unchanged.

Any failure is a falsifier.

10 Discussion

10.1 The two spaces

The state space X and the decision space $\mathcal{D}$ are distinct objects. The passage from one to the other is a quotient by task-indistinguishability. This quotient is not geometric: it is induced by the governed problem. A computation that acts on X without respecting $\sim$ wastes resources on distinctions the task cannot use. A computation that acts on $\mathcal{D}$ directly is geometry-native.

10.2 Quotient-consistency and its failure

The quotient-consistency theorem is the central result. It says that a transformation is well-defined on decision classes exactly when it respects the task equivalence relation. Failure of congruence yields ambiguity: the induced map depends on the choice of representative.

This is the formal content of the abstract's claim that failure of congruence yields ambiguity in the decision semantics. The ambiguity is not a modeling artifact: it is a mathematical obstruction.

10.3 Lossless compression

The lossless compression criterion is exact: a compression is lossless for a governed decision contract iff every fiber is contained in a task-equivalence class. The decision quotient is the finest lossless compression. Any coarser compression destroys a decision-relevant distinction.

This is the formal content of the abstract's claim that representation reduction necessarily destroys a distinction required for feasibility, identity, or certification when the reduction is not finer than the task equivalence.

10.4 Finite resources

The budget vector has six components. Each corresponds to a distinct resource demand. Omitting a component makes the certified output incommensurable with computations that respect the full vector.

Geometry-native computation has a budget advantage: it does not spend search budget on task-irrelevant distinctions. This is the formal reason geometry-native computation is efficient.

10.5 Randomization

Randomization generates candidates. It does not alter the objective, the feasible set, or the certification relation. This is a formal property of randomized candidate generators, not an empirical observation.

10.6 Relation to the constitutional framework

The two papers are complementary. The first establishes what states are permitted. The second establishes what states are distinguished. Together they provide a complete account of governed decision: admissibility selects the feasible set, and the task quotient selects the decision space. A geometry-native computation acts on both.

10.7 Limitations

The framework does not address:

- How to compute the task equivalence in practice.
- How to choose the protected invariants.
- How to allocate the budget vector.
- How to design a randomized candidate generator.

These are domain-specific questions. The framework provides the structure within which they can be posed.

10.8 Domain neutrality

The framework applies to any system with:

- a state space,
- a task structure,
- a governed decision contract,
- a metric or discernability form,
- admissible transformations,
- finite resources.

This includes adaptive computational systems, governance systems, biological regulatory networks, physical control systems, and any system that distinguishes states by task rather than by geometry.

11 Conclusion

We have formalized the passage from a state space to a decision space. The decision space is the quotient by task-indistinguishability. Geometry-native computation is computation whose admissible transformations act on this quotient. The quotient-consistency theorem shows that a transformation induces a well-defined map on decision classes exactly when it respects the task equivalence relation. Lossless compression is characterized by fibers contained in task-equivalence classes, and the decision quotient is the finest such compression. Finite computation is modeled through a budget vector covering representation, search, computation, verification, proof, and stopping. Randomization generates candidates but does not alter the objective, feasible set, or certification relation.

The framework is falsifiable by structural counterexamples and is domain-neutral. Its purpose is to make the state-versus-decision distinction precise and to provide the geometry on which governed computation can act.

References

[1] K. J. Arrow. *Social Choice and Individual Values*. Wiley, 1951.

[2] S. Awodey. *Category Theory*. Oxford University Press, 2nd edition, 2010.

[3] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[4] S. Mac Lane. *Categories for the Working Mathematician*. Springer, 2nd edition, 1998.

[5] K. R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.